Covariance

1.Definition Suppose X and Y are random variables, then the covariance of the r.v. X and Y is

$$Cov(X,Y) = E\left[(X-\mu_X)(Y-\mu_Y)\right]$$

where $\mu_X = E(X)$ and $\mu_Y = E(Y)$ exist.

when X = Y, $Cov(X,Y) = Var(X)$,
That's means variance is a special case of covariance.

- The variance of a single r.v. has been looked upon as a measure of dispersion of the distribution of the r.v.
- Some motivation will be given subsequently to the effect that the **Cov(X,Y)** may be thought of as a measure of the degree to which **X** and **Y** tend to increase or decrease simultaneously when **Cov(X, Y) > 0** or to move toward opposite directions when **Cov(X, Y) < 0.**

Covariance

2. A formula for calculating covariance

$$Cov(X,Y) = E(XY) - E(X)E(Y)$$

Prove:

$$Cov(X,Y) = E\{[X - E(X)][(Y - E(Y)]\}$$
$$= E(XY) - E(X)E(Y) - \cancel{E(Y)E(X)} + \cancel{E(X)E(Y)}$$
$$= E(XY) - E(X)E(Y)$$

Obviously,

if X and Y are independent to each other, then ***Cov(X, Y)= 0***

Covariance

New Question:

The value of the covariance reflects the relationship between X and Y to a certain extent, but it is also affected by measurement value and its units of X and Y.

$$Cov(X,Y) = E\{[X - E(X)][(Y - E(Y)]\}$$

$$Cov(aX, bY) = abCov(X,Y)$$

$$[Cov(X,Y)]^2 = Var(X)Var(Y)$$

Correlation coefficient

1. Definition The correlation coefficient of the r.v. X and Y is

$$\rho_{XY} = \rho(X,Y) = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

$$= \frac{Cov(X,Y)}{\sqrt{Var(X)} \cdot \sqrt{Var(Y)}}$$

where $\mu_X = E(X), \mu_Y = E(Y), \sigma_X = \sqrt{Var(X)}$ and $\sigma_Y = \sqrt{Var(Y)}$ exist.

2. Definition The r.v. X and Y are said to be uncorrelated if $\rho_{XY} = 0$.

Correlation coefficient

3. Theorem Let ρ_{XY} be the correlation coefficient of the random variables X and Y, then

(1) $\rho_{XY}^2 \leq 1$ or $|\rho_{XY}| \leq 1$;

(2) $\rho_{XY}^2 = 1$ or $|\rho_{XY}| = 1$

iff there are two constant a, b such that $Y = a + bX, (b \neq 0)$.

$P(Y = aX + b) = 1$

X and Y are linearly related with probability 1

Correlation coefficient

$$Cov(X,Y) = E\{[X - E(X)][(Y - E(Y)]\}$$
$$= E(XY) - E(X)E(Y)$$

X and Y are independent $\Rightarrow$ X and Y are uncorrelated

$\not\Leftarrow$

$E(XY) = E(X)E(Y)$ $\qquad$ $\rho_{XY} = 0$

- When X and Y are independent of each other, $\rho_{XY} = 0$, but its inverse is not true.
- $\rho_{XY} = 0$ does not means that X and Y are independent of each other.

Correlation coefficient

- When $\rho_{XY} = 0$, X and Y are uncorrelated.

  if X and Y are independent, then X uncorrelated to Y; but the opposite is not true.

- Only if in the following situations, independent and uncorrelated are equivalent.

  If (X, Y) obeys two-dimensional normal distribution, then:

  X and Y are independent $\Leftrightarrow$ X and Y are uncorrelated.

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Chapter 3 Numerical characteristics of random variables

Section 4 Moments and Covariance Matrix

- Moments
- Covariance matrix

Moments

Moments is a more extensive numerical feature of random variables, the mathematical expectation and variance introduced above are all moments.

Moments

Note:

- The mathematical expectation E(X) is the first-order origin moment of the random variable X;
- The variance D(X) is the second-order central moment of the random variable X;
- Covariance *Cov*(X,Y) is the second-order mixed central moment of random variables X and Y.

Covariance Matrix

Definition: The matrix

$$C = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix}$$

is the so called covariance matrix of two random variables X and Y if

$$c_{11} = E\left[(X - E(X))^2\right]$$

$$c_{12} = E[(X - E(X))(Y - E(Y))]$$

$$c_{21} = E[(Y - E(Y))(X - E(X))]$$

and

$$c_{22} = E\left[(Y - E(Y))^2\right].$$

Covariance Matrix

Similarly, the covariance matrix of *n*-dimensional r.v (X_1, X_2, …, X_n) can be defined.

Definition: The covariance matrix of n dimensional random variables $X_1, X_2, \cdots, X_n$ is defined as

$$C = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{mn} \end{pmatrix}$$

where

$$c_{ij} = \mathrm{Cov}\left(X_i, X_j\right)$$
$$= E\left[\left(X_i - E\left(X_i\right)\right)\left(X_j - E\left(X_j\right)\right)\right]$$
$$i, j = 1, 2, \cdots, n$$

Obviously the covariance matrix is symmetric.

Covariance Matrix

n-dimensional normal distribution

(1) Two-dimensional normal distribution

Let $X = (X_1, X_2)$ be a r.v. follows a two dimensional normal distribution $N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$ $(|\rho| < 1)$, then the covariance matrix of X is

$$C = \begin{pmatrix} \sigma_1^2 & \sigma_1\sigma_2\rho \\ \sigma_1\sigma_2\rho & \sigma_2^2 \end{pmatrix}$$

and

$$|C| = \sigma_1^2\sigma_2^2(1-\rho^2)$$

Covariance Matrix

(1) Two-dimensional normal distribution

If $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ and $\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$, it's easy to verify

$$(x-\mu)^T C^{-1}(x-\mu) =$$

$$\frac{1}{|C|}(x_1-\mu_1, x_2-\mu_2)\begin{pmatrix} \sigma_1^2 & -\sigma_1\sigma_2\rho \\ -\sigma_1\sigma_2\rho & \sigma_2^2 \end{pmatrix}\begin{pmatrix} x_1-\mu_1 \\ x_2-\mu_2 \end{pmatrix}$$

$$= \frac{1}{1-\rho^2}\left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]$$

Then the pdf of $X = (X_1, X_2)$ can be written as

$$f(x_1, x_2) = \frac{1}{2\pi|C|^{1/2}} e^{-\frac{1}{2}(x-\mu)^T C^{-1}(x-\mu)}$$

It's more simplify than the expression earlier seen.

Covariance Matrix

(2) *n*-dimensional normal distribution

If $X = (X_1, X_2, \cdots, X_n)$ is an n dimensional r.v., then its covariance matrix is

$$C = (c_{ij})_{n\times n} \quad i, j = 1, 2, \cdots, n$$

Let $x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ and $\mu = \begin{pmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_n) \end{pmatrix}$, then the pdf of $n-$ dimensional normal (Gaussian) distribution r.v. $X = (X_1, X_2, \cdots, X_n)$ is

$$f(X) = f(x_1, x_2, \cdots, x_n) = \frac{1}{(2\pi)^{n/2}|C|^{1/2}} e^{-\frac{1}{2}(x-\mu)^T C^{-1}(x-\mu)}$$

denoted as $X \sim N(\mu, C)$. You will obtain that $n-$ dimensional normal (Gaussian) distribution has many applications in theory and reality in the following chapters.

Covariance Matrix

➢ Four important properties of *n*-variable normal distribution

1. The necessary and sufficient condition for $\boldsymbol{X} = (\boldsymbol{X_1}, \boldsymbol{X_2}, \ldots, \boldsymbol{X_n})$ to obey n-dimensional normal distribution is that :
   For all real numbers $a_1, a_2, \ldots, a_n$ (not all zero) , any linear combination of $(X_1, X_2, \ldots, X_n)$: $\boldsymbol{a_1X_1} + \boldsymbol{a_2X_2} + \ldots + \boldsymbol{a_nX_n}$ obeys one-dimensional normal distribution.

2. If $\boldsymbol{X} = (\boldsymbol{X_1}, \boldsymbol{X_2}, \ldots, \boldsymbol{X_n})$ obeys *n*-dimensional normal distribution, $\boldsymbol{Y_1}, \boldsymbol{Y_2}, \ldots, \boldsymbol{Y_k}$ are linear functions of X_j (j = 1, 2,…, n), then $(Y_1, Y_2, \ldots, Y_k)$ also obeys the multidimensional normal distribution.

This property is called the linear transformation invariance of normal variables.

Covariance Matrix

- Four important properties of *n*-variable normal distribution

3. If $X = (X_1, X_2, \ldots, X_n)$ obeys the n-dimensional normal distribution, then each of its components X_j (j = 1, 2,…, n) obeys the normal distribution; conversely, if $X_1, X_2, \ldots, X_n$ all obey the normal distribution and independent to each other, then $(X_1, X_2, \ldots, X_n)$ obeys *n*-dimensional normal distribution.

4. Assuming $(X_1, X_2, \ldots, X_n)$ obeys n-dimensional normal distribution, then the sentence "$X_1, X_2, \ldots, X_n$ are mutually independent" is equivalent with the sentence "$X_1, X_2, \ldots, X_n$ are uncorrelated".

This four properties will be often used in the subsequent courses of "random process" and "mathematical statistics".

Covariance Matrix

Example: Assuming the r.v *X* and *Y* are independent to each other, and X ~ N (1, 2), Y ~ N (0, 1).

Calculate: the probability density of Z = 2X – Y + 3

Answer:

∵ X ~ N (1, 2), Y ~ N (0, 1), and X and Y are independent.

∴ the joint distribution of X and Y is a normal distribution;
and any linear combination of X and Y obeys normally distributed.

Thus, Z ~ N (E(Z), D(Z))

and E(Z) = 2E(X) - E(Y) + 3 = 2 + 3 = 5
D(Z) = 4D(X) + D(Y) = 8 +1 = 9

So, **Z ~ N (5, 3^2)** therefore, the probability density of Z is:

$$f_Z(z) = \frac{1}{3\sqrt{2\pi}} e^{-\frac{(z-5)^2}{18}} \qquad -\infty < z < \infty$$